

TPL-08 · FREE TEMPLATES

Estimate at completion scenario comparison

One input block, four methods side by side, and a recommendation field that forces you to say why



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PROJECT CONTROLS INSTITUTE GLOBAL, INC.

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— Summary

Four standard ways of computing an estimate at completion will give four different answers from the same inputs, because each encodes a different assumption about whether the variance suffered so far will happen again. This template puts them side by side from one input block, computes variance at completion and the to-complete performance index for each, names the assumption each one encodes, and requires the user to record which was selected and why.

— About this document

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Contents

1. When to use this	4
2. How to complete it	4
3. The template	5
3.1 Input block	5
3.2 The method comparison table	6
3.3 To-complete performance index to budget at completion	7
3.4 The three arithmetic self-checks	8
3.5 The recommendation block	9
4. Worked fragment	9
5. Common mistakes	11
6. Adapting it	12
7. Completion checklist	12
Related	13
Sources and standards	13
Status and version	13

Estimate at completion scenario comparison

In one paragraph. Four standard ways of computing an estimate at completion will give four different answers from the same inputs, because each encodes a different assumption about whether the variance suffered so far will happen again. This template puts them side by side from one input block, computes variance at completion and the to-complete performance index for each, names the assumption each one encodes, and requires the user to record which was selected and why.

Who this is for. Cost engineers and project controls managers who prepare the forecast; project directors and sponsors who approve it; and anyone who has been asked why the estimate at completion moved and needs an answer better than "the index changed".

— 1. When to use this

Produce the comparison every period in which a forecast is reported, immediately after the earned value sheet (TPL-07) closes. Producing a single estimate at completion by one method and reporting it as the forecast conceals the judgement that was made; producing four and selecting one makes the judgement visible and reviewable.

Three moments make it more than a routine:

- **The first period in which the cost performance index departs materially from one.** That is the period in which the project decides, usually without noticing, whether it believes the variance is structural or a one-off. The sheet forces the decision to be recorded.
- **Before a request for additional funding or a contingency release.** The forecast is the basis of the request, and the spread between methods is the honest expression of its uncertainty.
- **When the selected method changes.** A change of method moves the forecast without anything happening on site. It must be disclosed as a method change, with the previous method's answer shown alongside for one period.

Do not use index-based methods early. Below roughly fifteen to twenty per cent complete the indices are ratios of small numbers, and extrapolating them across the remaining eighty per cent produces a forecast with a wide error and a confident appearance. Early in a project, the bottom-up method is the only one worth reporting.

— 2. How to complete it

Enter the input block once. Budget at completion, planned value, earned value and actual cost, all to the same data date and all from the earned value sheet. Everything else on this sheet is computed from them. The definitions and interpretation of the indices belong to TPL-07 and are not restated here; they are computed in the input block so that this sheet stands alone.

State which budget at completion you are using. The performance measurement baseline — distributed control-account budget — is the correct base for an index-based forecast, because it is the only budget against which earned value has been measured. If contingency sits inside your budget at completion, take it out for this calculation and show the remaining contingency separately below the forecast. A forecast that quietly absorbs contingency has removed a decision from the person entitled to make it.

Compute the bottom-up estimate to complete separately, and properly. It is the only method that is not an extrapolation, and it is the only one that can see a change in the work ahead. Ask each control account manager for a re-estimate of the remaining work at current rates, with the same estimate basis discipline you would apply to a change request (TPL-04): quantity, rate, source, and the assumptions the number depends on. A bottom-up figure assembled by scaling the remaining budget is not a bottom-up figure.

Read the to-complete performance index as a credibility test, not as a target. It states the cost performance the remaining work must achieve for a given endpoint to be met. Compare it with the cost performance index actually achieved. When the required performance is materially better than anything the project has yet demonstrated, the endpoint is an aspiration and the sheet says so in one number.

Use the three arithmetic self-checks in §3.4. Each index-based method has a to-complete performance index that must come out to a specific value. If yours does not, the sheet has an error in it.

Compute at full precision, round for display. Using a cost performance index rounded to three decimal places rather than the underlying ratio moves the first method's answer in the worked example by CU 2 thousand. On a larger project the same error is material.

Fill the recommendation block before the forecast leaves the room. The methods do not choose between themselves. The block requires the selected method, the reason it was selected over the others, what would change the choice, and who challenged it.

Using the tables. Copy a table block, paste into a single spreadsheet column, split on the pipe character, and delete the alignment row.

— 3. The template

3.1 Input block

Hold the inputs in a single column so every method references the same cells.

Cell	Field	Type	Entry rule
B1	Project / control account	Text	The level at which this forecast is made
B2	Data date	Date	Must match the earned value sheet
B3	Budget at completion (BAC)	Input	Distributed performance measurement baseline
B4	Planned value (PV)	Input	Cumulative to the data date
B5	Earned value (EV)	Input	Cumulative to the data date
B6	Actual cost (AC)	Input	Cumulative to the data date, including accruals
B7	Cost performance index (CPI)	Calculated	=IF(N(\$B\$6)=0, "", \$B\$5/\$B\$6)
B8	Schedule performance index (SPI)	Calculated	=IF(N(\$B\$4)=0, "", \$B\$5/\$B\$4)
B9	Bottom-up estimate to complete	Input	Re-estimate of the remaining work, assembled from the control accounts
B10	To-complete performance index to BAC	Calculated	See §3.3
B11	Remaining contingency	Input	From the change register (TPL-04)
B12	Currency, scale and basis	Text	e.g. CU thousands, nominal, no escalation

3.2 The method comparison table

Four rows, one per method. Data rows 15 to 18.

Col	Field	Type	Definition
A	Method	Text	
B	Assumption this method encodes	Text	Pre-filled below; do not paraphrase it away
C	Estimate at completion	Calculated	
D	Variance at completion	Calculated	
E	Variance at completion %	Calculated	
F	To-complete performance index to this estimate	Calculated	
G	Evidence supporting this assumption	Text	What is true about this project that makes it right
H	Evidence against	Text	What is true about this project that makes it wrong

Method 1 — Budget at completion divided by the cost performance index. In words: the remaining work will be delivered at the same cost efficiency as the work already done. Formula: $EAC = BAC \div CPI$. Spreadsheet, written so it never divides by a rounded or blank index:

```
=IF(OR(N($B$5)=0,N($B$6)=0), "", $B$3*$B$6/$B$5)
```

Since the cost performance index is earned value divided by actual cost, dividing budget at completion by it is the same as multiplying budget at completion by actual cost and dividing by earned value. The second form evaluates at full precision and guards both denominators.

Method 2 — Actual cost plus the remaining budget. In words: the variance to date was a one-off, and the remaining work will be delivered at the budgeted rate. Formula: $EAC = AC + (BAC - EV)$. Spreadsheet:

```
=IF(OR($B$3="", $B$5="", $B$6=""), "", $B$6+($B$3-$B$5))
```

Method 3 — Actual cost plus the remaining budget divided by the cost and schedule indices compounded. In words: the remaining work will be affected both by the cost efficiency achieved so far and by the schedule pressure implied by working behind plan. Formula: $EAC = AC + (BAC - EV) \div (CPI \times SPI)$. Spreadsheet:

```
=IF(OR(N($B$4)=0, N($B$5)=0, N($B$6)=0), "", $B$6+($B$3-$B$5)/((($B$5/$B$6)*($B$5/$B$4)))
```

All three denominators are guarded: the two indices need non-zero actual cost and planned value, and the compound product needs non-zero earned value.

Method 4 — Actual cost plus a bottom-up estimate to complete. In words: historical performance does not describe the work remaining, so the remaining work has been re-estimated. Formula: $EAC = AC + ETC$. Spreadsheet:

```
=IF($B$9="", "", $B$6+$B$9)
```

Variance at completion (column D). In words: budget at completion less the estimate at completion. Negative means a forecast overrun.

```
=IF(NOT(ISNUMBER(C15)), "", $B$3-C15)
```

Variance at completion per cent (column E). In words: variance at completion as a proportion of budget at completion.

```
=IF(OR(NOT(ISNUMBER(D15)), N($B$3)=0), "", D15/$B$3)
```

To-complete performance index to this estimate (column F). In words: the cost performance the remaining work must achieve for this estimate to be met — remaining budgeted work divided by remaining forecast money.

```
=IF(OR(NOT(ISNUMBER(C15)), C15=$B$6), "", ($B$3-$B$5)/(C15-$B$6))
```

The `ISNUMBER` test is what makes the guard safe: a blank estimate cell is text, and subtracting actual cost from it would produce an error that propagates across the row.

3.3 To-complete performance index to budget at completion

In words: the cost performance the remaining work must achieve to finish within the original budget. Formula: $TCPI \text{ to } BAC = (BAC - EV) \div (BAC - AC)$. Spreadsheet, in cell B10:

```
=IF(OR($B$3="", $B$3=$B$6), "", ($B$3-$B$5)/($B$3-$B$6))
```

Where actual cost has already reached or passed budget at completion the denominator is zero or negative and the measure stops being meaningful — the guard returns blank, and the honest report is that the budget cannot be met, not a negative index.

Compare **B10** with **B7**. The ratio of the two is how much better than achieved performance the remaining work must run:

```
=IF(OR(N($B$7)=0, $B$10=""), "", $B$10/$B$7-1)
```

3.4 The three arithmetic self-checks

Each index-based method implies a specific to-complete performance index. These are identities, not coincidences, and they are the fastest way to find an error in the sheet.

Method	Its to-complete performance index must equal	Why
1 — $BAC \div CPI$	the cost performance index	Substituting $EAC = BAC \times AC \div EV$ gives $EAC - AC = AC \times (BAC - EV) \div EV$, so the index reduces to $EV \div AC$
2 — $AC + (BAC - EV)$	exactly 1.000	$EAC - AC = BAC - EV$, so numerator and denominator are identical
3 — $AC + (BAC - EV) \div (CPI \times SPI)$	the cost index multiplied by the schedule index	$EAC - AC = (BAC - EV) \div (CPI \times SPI)$, so the ratio is $CPI \times SPI$

Method 4 has no identity, because its estimate to complete is entered rather than derived. That is the point of it.

3.5 The recommendation block

Field	Entry
Method selected	
Why this method rather than the other three	Required. Name the cause of the variance to date and say whether it applies to the work remaining. A reason that would be true on any project is not a reason.
Evidence relied on	Estimate basis, quotations, productivity records, subcontract position
Previous period's method and estimate	If the method has changed, both are reported for one period
Range reported	Lowest and highest of the four, as the honest expression of uncertainty
What would change this choice	The specific observation that would move the forecast next period
Remaining contingency	From the change register
Variance at completion against remaining contingency	Whether the forecast can be absorbed
Prepared by / date	
Challenged by / date	A second person, not the preparer
Approved by / date	
Next review	

— 4. Worked fragment

Illustrative figures. The same fictional facility upgrade project as [TPL-07](#), at the same cut-off.

- **Currency and scale:** generic currency units (CU), thousands.
- **Basis:** nominal, single currency, no escalation.
- **Data date:** 31 May 2026.
- **Budget at completion used:** CU 8,000 thousand, the distributed performance measurement baseline. The contingency reserve of CU 300 thousand sits outside it and is reported separately below.
- **Rounding:** amounts to the nearest CU thousand; indices to three decimal places; percentages to one decimal place, half away from zero. Adverse figures in parentheses. All arithmetic performed at full precision.

Input block.

Field	Value
Budget at completion	8,000
Planned value	4,000
Earned value	3,800
Actual cost	3,980
Cost performance index	0.955
Schedule performance index	0.950
Bottom-up estimate to complete	4,450
To-complete performance index to budget at completion	1.045
Remaining contingency	300

Indices verified: $CPI = 3,800 \div 3,980 = 0.955$ (0.954774 unrounded); $SPI = 3,800 \div 4,000 = 0.950$; $TCPI \text{ to BAC} = (8,000 - 3,800) \div (8,000 - 3,980) = 4,200 \div 4,020 = 1.045$.

Method comparison.

Method	Assumption encoded	EAC	VAC	VAC %	TCPI to this EAC
1 — $BAC \div CPI$	Cost efficiency achieved to date continues across all remaining work	8,379	(379)	(4.7 %)	0.955
2 — $AC + (BAC - EV)$	The variance to date was a one-off; remaining work runs at budget	8,180	(180)	(2.3 %)	1.000
3 — $AC + (BAC - EV) \div (CPI \times SPI)$	Remaining work is affected by both cost efficiency and schedule pressure	8,610	(610)	(7.6 %)	0.907
4 — $AC + \text{bottom-up ETC}$	History does not describe the work ahead; it has been re-estimated	8,430	(430)	(5.4 %)	0.944

Verification of each method.

Method 1: $8,000 \div 0.954774 = 8,378.95$, so CU 8,379 thousand. Equivalently $8,000 \times 3,980 \div 3,800 = 8,378.95$. Variance at completion = $8,000 - 8,378.95 = (378.95)$, shown as (379); $(378.95) \div 8,000 = (4.7 \%)$. Using the rounded index of 0.955 instead would give $8,000 \div 0.955 = 8,376.96$, CU 2 thousand adrift — which is why the formula in §3.2 avoids the index cell.

Method 2: $3,980 + (8,000 - 3,800) = 3,980 + 4,200 = 8,180$. Variance at completion = $8,000 - 8,180 = (180)$; $(180) \div 8,000 = (2.25 \%)$, displayed as (2.3 %). Note that this variance at completion equals the cost variance in [TPL-07](#), which it must: the method assumes no further variance is incurred.

Method 3: $CPI \times SPI = 0.954774 \times 0.950 = 0.907035$. Remaining budgeted work $\div$ that product = $4,200 \div 0.907035 = 4,630.47$. Estimate at completion = $3,980 + 4,630.47 = 8,610.47$, so CU 8,610 thousand. Variance at completion = $8,000 - 8,610.47 = (610.47)$, shown as (610); $(610.47) \div 8,000 = (7.6 \%)$.

Method 4: $3,980 + 4,450 = 8,430$. Variance at completion = $8,000 - 8,430 = (430)$; $(430) \div 8,000 = (5.4 \%)$. The bottom-up estimate to complete of 4,450 sits CU 250 thousand — 6.0 per cent — above the remaining budgeted work of 4,200.

Verification of the self-checks. Method 1: $4,200 \div (8,378.95 - 3,980) = 4,200 \div 4,398.95 = 0.955$, equal to the cost performance index. Method 2: $4,200 \div (8,180 - 3,980) = 4,200 \div 4,200 = 1.000$. Method 3: $4,200 \div (8,610.47 - 3,980) = 4,200 \div 4,630.47 = 0.907$, equal to $CPI \times SPI$. All three identities hold, so the sheet is arithmetically sound.

The credibility test. Finishing within the original budget requires a to-complete performance index of 1.045 against a cost performance index of 0.955 achieved so far — a ratio of 1.094, meaning the remaining work would have to be delivered about **nine per cent more efficiently than anything the project has yet demonstrated**, having so far run about five per cent worse than budget. Nothing in the record supports that, so budget at completion is not a forecast; it is a target.

The range. The four methods span CU 8,180 to CU 8,610 thousand, a spread of CU 430 thousand, or 5.4 per cent of budget at completion. That spread is the honest statement of forecast uncertainty at this data date, and it belongs in the monthly report next to whichever single figure is selected.

Recommendation, as it would be written.

Method selected: Method 4, bottom-up, at CU 8,430 thousand.

Why this method rather than the other three. The cost variance to date is concentrated in CA-1000 and has two identified causes: piling productivity below the estimate basis, and rework at the pile cap interface. The piling cause is structural and will persist across the remaining 132 piles, so Method 2 — which assumes the variance was a one-off — is rejected. The rework cause has been closed by a revised inspection sequence and will not recur, so Method 1, which projects the whole variance to date across all remaining work including the mechanical account that is currently ahead, overstates it. Method 3 additionally loads the schedule performance index onto the forecast; the schedule variance here arises in CA-3000, whose work is largely still ahead and is not resource-constrained, so compounding is not supported. The control account managers have re-estimated the remaining work at CU 4,450 thousand against a remaining budget of CU 4,200 thousand, with the piling shortfall priced at current measured rates.

What would change this choice. If piling output over June does not recover to the re-estimated rate, the bottom-up basis fails and Method 1 becomes the defensible forecast. If mechanical installation begins to consume float, Method 3 becomes relevant.

Funding position. Forecast variance at completion of CU (430) thousand against remaining contingency of CU 300 thousand — a shortfall of CU 130 thousand — before BCR-014, which itself requests CU 295 thousand from that contingency and is under review (TPL-04 \$4.1). If BCR-014 is approved, CU 5 thousand of contingency remains against a forecast overrun of CU 430 thousand. That is the decision the sponsor is being asked for, and it belongs on page one of the monthly report.

— 5. Common mistakes

Reporting one method as though it were the answer. The single figure hides the assumption. Report the selected figure with the range and the reason.

Using an index-based method too early. Below roughly fifteen to twenty per cent complete the extrapolation is built on a ratio of small numbers, and its confident appearance is the dangerous part.

Using a rounded index in the formula. Small on this example, material on a large project, and always avoidable by referencing the input cells.

Applying the compound method by default. Multiplying by the schedule performance index assumes that being behind schedule will cost money on the remaining work. Sometimes it will — acceleration, out-of-sequence working, extended preliminaries. Sometimes the schedule variance sits in work that is not resource-constrained and costs nothing extra. Compounding without that argument is not conservatism; it is an unexamined assumption dressed as prudence.

A "bottom-up" estimate that is the remaining budget with a factor applied. That is Method 1 with extra steps, and it inherits none of the diagnostic value.

Treating the to-complete performance index as a target. It is a statement of what is required, not an instruction. Circulating it as a productivity goal is how a forecast becomes a negotiation.

Forgetting that a contingency release changes budget at completion. When a contingency release moves budget into a control account, budget at completion for measurement rises and every index, variance and forecast is computed on the new base. Recompute; do not carry the previous period's forecast forward.

Changing method quietly. The forecast moves and nothing happened. Disclose the change, show both answers for one period, and say what caused the change of view.

Comparing the forecast overrun with total contingency rather than remaining contingency, or ignoring pending change requests against it. The relevant number is what is left after everything already approved and everything currently in the approval queue.

— 6. Adapting it

Safe to change. Running the sheet at control-account level as well as at project level, which is usually more informative — a project-level forecast averages away the account where the problem is. Adding methods: a weighted compound variant that blends the cost and schedule indices in some other proportion is legitimate provided the weighting is stated and justified on the sheet rather than inherited from a spreadsheet somebody was given. Adding an optimistic and pessimistic bottom-up case to widen the range deliberately.

Change with care. Adding a probabilistic forecast from a cost risk analysis. It answers a different question — the distribution of outcomes rather than a point estimate under an assumption — and placing it in the same table invites readers to compare a percentile with a deterministic figure as though they were the same kind of number. If you add it, label it separately.

Do not remove. The assumption column, the to-complete performance index column, the range, and the recommendation block's "why this method rather than the others" field. Without those four the sheet becomes a menu of numbers from which the most convenient can be chosen.

Where the client mandates a method, compute theirs and report it as the contractual forecast, and keep this comparison as the internal position with the difference reconciled. Do not let a mandated method become the project's own view of itself.

— 7. Completion checklist

- [] Inputs match the earned value sheet exactly, to the same data date
- [] Budget at completion used is the distributed performance measurement baseline, stated as such

- [] Contingency shown separately, not absorbed into the forecast
- [] All four methods computed from the same input cells
- [] Every division guarded, and every estimate cell tested with `ISNUMBER` before subtraction
- [] Formulas reference unrounded inputs, not displayed indices
- [] The three arithmetic self-checks in §3.4 all hold
- [] To-complete performance index to budget at completion computed and compared with the achieved index
- [] Bottom-up estimate to complete assembled from control accounts with a stated estimate basis
- [] Assumption column completed for every method, with evidence for and against
- [] Method selected, with a reason specific to this project's variance causes
- [] Range reported alongside the selected figure
- [] Variance at completion compared with remaining contingency after pending changes
- [] Forecast challenged by a second person and approved, both named and dated
- [] Any change of method disclosed, with the previous method's answer shown for one period

— Related

- [TPL-07 – Earned value calculation sheet](#) — the source of every input, and where the indices are defined
- [TPL-04 – Baseline change request](#) — the contingency position the forecast must be read against
- [TPL-06 – Monthly project controls report](#) — where the selected forecast and its range are reported
- [BPG-09 – Estimate at completion – choosing and defending a method](#) — the judgement in full
- [BPG-10 – Contingency and management reserve](#) — how a forecast overrun and contingency relate

— Sources and standards

- PCI Master Formula Sheet ([docs/downloads/master-formula-sheet.md](#)), August 2026: the estimate at completion, variance at completion and to-complete performance index formulas used throughout. Published under the credential's retired code; the credential is PCL-AI.

The four methods are common to established earned value practice and are stated here in the Institute's own words. No third-party sheet, table or wording is reproduced. The identities in §3.4 are derived from the formulas themselves and are verified in §4.

— Status and version

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